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Original Article

Therapeutic effects of the total lignans from *Vitex negundo* seeds on collagen-induced arthritis in rats

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A R T I C L E I N F O

Keywords:
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A B S T R A C T

Background: The seeds of *Vitex negundo*, with rich lignans metabolites, have been widely used as a traditional Chinese medicine and Ayurvedic herbal medicine for the treatment of rheumatism and joint inflammation. The total lignans of *Vitex negundo* seeds (TOV) were suggested to play an important role in the treatment of arthritis.
Purpose: The aim of the study was designed to investigate the anti-arthritic effects of TOV on collagen-induced arthritis (CIA) in rats as well as its possible mechanisms.
Methods: TOV was prepared by combined macroporous resin and polyamide column chromatography, and constituents of TOV were analyzed by HPLC. CIA model in rats was established by immunization with chicken type II collagen and then the rats were intragastrically administrated with TOV for 30 days. Rat arthritis was evaluated by measurements of hind paw edema, arthritis index score, weight growth and indices of thymus and spleen, and by histological examination. Levels of serum MMP-2, MMP-3, MMP-9, IL-1*β*, IL-6, IL-8, IL-10, IL-17A and TNF-α were also examined. In addition, the expression of COX-2, iNOS and IκB, *p*–IκB in synovial tissues was evaluated by western blotting. The analgesic and anti-inflammatory effects of TOV were also evaluated in acetic acid-induced writhing and xylene-induced ear edema in mice, respectively. In addition, acute toxicity test was employed to preliminarily assess the safety of TOV.
Results: TOV significantly inhibited the paw edema and decreased the arthritis index, with no influence on the body weight and the indices of thymus and spleen of CIA rats. Meanwhile, TOV dose-dependently reduced the infiltration of inflammatory cells, synovial hyperplasia and attenuated cartilage damage. Additionally, the serum levels of IL-1β, IL-6, IL-8, IL-17A, TNF-α, MMP-3 and MMP-9 were markedly decreased, while the level of serum IL-10 was increased in TOV-treated rats. The significant reduction of the expression of COX-2, iNOS and *p*-IκB and the notable increase of IκB in synovial tissues were also observed in TOV-treated animals. TOV also significantly inhibited acetic acid-induced writhing and decreased xylene-induced ear edema in mice. Finally, the maximal tolerable dose (MTD) of TOV was determined to be 16.0 g/kg.
Conclusion: These results suggest that TOV has significant anti-arthritic effects on collagen-induced arthritis in rats, which may be attributed to the inhibition of the levels of IL-1β, IL-6, IL-8, IL-17A, TNF-α, MMP-3 and MMP-

Abbreviations: CFA, complete Freund's adjuvant; CIA, collagen-induced arthritis; CII, collagen type II; COX-2, cyclooxygenase-2; DMARDs, disease-modifying antirheumatic drugs; DXM, dexamethason; IκB, inhibitor of NF-κB; IFA, incompere Freund's adjuvant; IL-1β, interleukin-1β; IL-6, interleukin-6; IL-8, interleukin-8; IL-10, interleukin-10; IL-17A, interleukin-17A; iNOS, inducible nitric oxide synthase; MMP-2, matrixmetalloproteinases-2; MMP-3, matrixmetalloproteinases-3; MMP-9, matrixmetalloproteinases-9; MTD, maximal tolerable dose; NF-κB, nuclear factor-κB; NSAIDs, non-steroidal anti-inflammatory drugs; RA, rheumatoid arthritis; TNF-*α*, tumor necrosis factor-α; TOV, total lignans of *Vitex negundo* seeds; VNL, 6-hydroxy-4-(4-hydroxy-3-methoxyphenyl)-3-hydroxymethyl-7-methoxy-3,4- dihydro-2-naphthaldehyde

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9, and the increase of IL-10 in serum as well as down-regulation of the protein expression of COX-2 and iNOS in synovial tissues via suppressing the phosphorylation and degradation of IκB. Due to its high efficacy and safety, TOV can be regarded as a promising drug candidate for rheumatoid arthritis treatment.

Introduction

Rheumatoid arthritis (RA) is one of the most prevalent autoimmune inflammatory diseases, which has a significant negative impact on the adult ability to perform daily activities, including work, exercise and household tasks, and substantially decreased health-related quality of life, with increased mortability (Pincus et al., 1984; Salaffi et al., 2009; Solomon et al., 2003). It primarily involves the joints, but should be considered as a syndrome that is associated with many extraarticular manifestations, such as rheumatoid nodules, pulmonary involvement or vasculitis (Scott et al., 2010) .

Although disease-modifying antirheumatic drugs (DMARDs, *e.g.* methotrexate) and non-steroidal anti-inflammatory drugs (NSAIDs, *e.g.* indomenthacin), as well as glucocorticoids, could alleviate the symptoms of RA patients, their clinical use is limited due to the continuous disease progression despite ongoing therapy and the emergence of side effects (Roubille et al., 2015; Smolen and Aletaha, 2015). Specifically, DMARDs target inflammation and by definition must reduce structural damage progression. However, the uses of DMARDs have been impeded by their potential of long-term side effects and toxicity (Kremer, 1999). NSAIDs, while reducing pain and stiffness and improving physical function, do not interfere with joint damage and are thus not disease modifying (Scott et al., 2010). Glucocorticoids offer rapid symptomatic and disease-modifying effects, but are also associated with serious long-term side-effects (Kirwan, 1995). A better understanding of the important role of pro-inflammatory cytokines in the pathophysiology of RA in recent times has vastly improved treatment options, as evidenced by the development of pro-inflammatory cytokine blockers (Furst and Emery, 2014; Siebert et al., 2015). However, these biological agents achieve disease remission in limited number of patients due to infectious adverse events, rebound of symptoms, or high treatment costs (Huscher et al., 2015; Lahiri and Dixon, 2015; Zampeli et al., 2015). Therefore, it is increasingly important and urgent to search for new therapeutic drugs with high efficiency and lower toxicity for RA treatment from the natural sources.

Collagen-induced arthritis (CIA) is one of the most common animal model of inflammatory arthritis for disease pathogenesis investigation and validation of potential therapeutic targets (Lorenzo et al., 2012). Both humoral and cellular immune mechanism are involved in the induction and mediation of CIA after immunization by collagen type II (CII) (Banda et al., 2002). In addition, the inflammatory progress mediated by anti-CII antibody, complement components and cytokines are also important in the cartilage and bone damage which is a characteristic of CIA (Banda et al., 2002). Therefore, CIA can be served as appropriate animal model with both clinical and histological similarities to human RA for evaluating the effects of anti-rheumatic drugs.

Vitex negundo (Verbenaceae) is one of the most important medicinal plants of the *Vitex* species. The seeds are frequently used in traditional Chinese medicine and Ayurveda for anti-inflammatory (Chawla et al., 1992; Zheng et al., 2009a; Zheng et al., 2010), analgesic (Zheng et al., 2009a), antitumor (Zhou et al., 2009) and antioxidant purposes (Zheng et al., 2009b). A variety of secondary metabolites, such as lignans (Ono et al., 2004; Zheng et al., 2009a), terperniods (Ono et al., 2004; Zheng et al., 2010), flavonoids and steriods (Chawla et al., 1992; Zhao et al., 2012), were reported from this medicinal seeds. Based on our previous investigation, the ethanolic extract of *V. negundo* seeds showed potent anti-inflammatory activity in the xylene-induced ear edema test (Zheng et al., 2009b) and anti-arthritic effects on complete Freund's adjuvant induced arthritis in rats (Zheng et al., 2014b). In addition, we found that phenylnaphthalene-type lignans were the main anti-inflammatory components in the extract of *V. negundo* seeds (Zheng et al., 2009a; Zheng et al., 2009b). Therefore, we suggested that the total lignans of *V. negundo* seeds (TOV) play an important role in its traditional use for the treatment of rheumatoid arthritis and the present study was thus conducted to prepare TOV from *V. negundo* seeds and investigate the anti-arthritic effects of TOV on collagen-induced arthritis (CIA) in rats as well as its possible mechanisms.

Materials and methods

Chemicals and reagents

Dexamethason (DXM) was purchased from Shanghai Xinyi Pharmaceutical Co., Ltd (Shanghai, China). Chicken type II collagen (CII) was bought from Xi'an Herbking Biotechnology Co., Ltd (Xi'an, China). Complete Freund's adjuvant (CFA) and Incompere Freund's adjuvant (IFA) were purchased from Sigma chemical Co., USA (Milwaukee, WI, USA). Elisa kits of IL-8, IL-10, IL-17A, MMP-2, MMP-3 and MMP-10 were obtained from NeoBioscience Technology Co., Ltd (Shenzhen, china), while IL-1β, IL-6 and TNF-α were acquired from eBiosciences Inc.,USA (San Die go, CA, USA). Antibodies against IκB, phospho-IκB (Ser23), COX2, iNOS, GAPDH and horseradish peroxidase (HRP)-conjugated anti-mouse IgG were purchased from Nanjing Enogene Biological Technology Co., Ltd (Nanjing, China).

Plant material

The seeds of *V. negundo* (Chinese name 'Huangjingzi') were collected from Jianyang city of Sichuan province, and were identified by Professor Lu-ping Qin, Second Military Medical University. A voucher specimen (NO. 2014-043) has been deposited in the Hebarium of Department of Pharmacognosy, School of pharmacy, Second Military Medical University.

Preparation of the total lignans of Vitex negundo seeds (TOV)

The seeds of *Vitex negundo* seeds (10.0 kg) were roughly powdered and extracted twice with 70% ethanol (10 ×) under reflux for 1 h each time. The solvent was evaporated under vacuum to afford the crude extract (0.5 kg), which was then subjected to polyamide column chromatography, eluted with distilled water and 35% ethanol consecutively to afford the targeted fraction. This fraction was then suspended in distilled water with the concentration of 1.5 mg/ml for use. The suspension was then chromatographed on a D101 macroporous resin column, washed with distilled water to remove polysaccharides and later eluted with 35% ethanol to give the eluent. Subsequently, the eluent was evaporated under vacuum, re-suspended in distilled water, and finally partitioned with ethyl acetate to obtain the ethyl acetate fraction. The ethyl acetate fraction was condensed and lyophilized by freeze-drying to afford TOV (20.0 g) and stored in –20 °C until usage.

HPLC analysis of TOV

The chromatographic analysis was performed on an Agilent 1200 HPLC system (Agilent Technologies, USA), equipped with a quaternary pump, a degasser, an automatic thermostatic column compartment, and a computer with a chemstation software program for HPLC data analysis. Separation was performed on a C18 (250 × 4.6 mm i.d.; 5 μm, Agilent, USA) column using a gradient elution at a flow rate of 1.0 ml/min. The mobile phase was composed of A: acetonitrile and B: 0.1% formic acid solution (v/v). The gradient program was set as follows: 0–18 min, 14.4% A in B; 18–30 min, 14.4% – 17.0% A in B, 30–45 min, 17.0% A in B; 45–75 min 17.0% – 43.0% A in B. VNL (6-hydroxy-4-(4-

hydroxy-3-methoxyphenyl)-3-hydroxymethyl-7-methoxy-3,4-dihydro-2-naphthaldehyde), vitedoin A, vitedoamine A and vitexdoin A were previously obtained in our lab and used as reference compounds to identified the main compounds in TOV. The purity of these four lignans used in our study was determined to be over 95%, according to the HPLC area normalization method in our previous study (Shu et al., 2016), which was further evidenced by our findings that the purity factors were within the scope of the threshold value and that the threshold curves (Fig. 1) did not intersect the purity curves (Da Costa et al, 2015).

Animals

Adult SD rats (SPF grade) weighing 180–220 g and ICR mice (SPF grade) weighing 18–20 g, with an equal proportion of males and females, were provided by Beijing Vital River Laboratory Animal Technology Co., Ltd (Animal lisence number: SCXK (Beijing) 2012-0001). The animals were fed with standard diet and water, and were kept with photoperiod (12:12 h) light-dark cycle and a temperature of 24 ± 2 °C. Animal experiments were conducted in accordance with international ethical guidelines and the National Institutes of Health Guide concerning the Care and Use of Laboratory Animals, and the experiments were carried out with the approval of the Animal Experimentation Ethics Committee of Second Military Medical University.

Induction of CIA and administration of TOV

All rats, except the normal group, were immunized intradermally in two sites at the base of the tail with 100 μg of chicken type II collagen (CII) as described previously (Stump et al., 2011). Briefly, on day 0, SD rats were given a primary immunization with equal volumes of chicken type II collagen, 4 mg/ml stock in 0.1 mM acetic acid, emulsified in Complete Freund's adjuvant (CFA). At 14 days post primary immunization, rats were rechallenged with CII emulsified in Incomplete Freund's adjuvant (IFA) intradermally. On day 21, rats were divided randomly into six groups with ten animals in each group as follows: group I (normal control), group II (CIA model), group III (DXM 0.05 mg/kg), group IV (TOV 10 mg/kg), group V (TOV 20 mg/kg) and group VI (TOV 40 mg/kg). DXM and TOV were orally administered from day 21 to 51. Normal and CIA model group rats were orally given an equal volume of vehicle in the same schedule. During the course of the experiment, the body weight and the volume of left and right hind paw of rats were measured every 3days. And the rats were assessed every three days for sign of arthritis between days 21 and 51. Individual paw was recorded on an ordinal scale of 1–4, as follows: 0 = unaffected and no sign of arthritis, 1 = 1 type of joint affected with erythema and mild swelling confined to the ankle joint and toes, 2 = 2 type of joint affected with erythema and mild swelling extending from the ankle joint to midfoot, 3 = 3 type of joint affected erythema and severe swelling extending from the ankle joint to metatarsal joints, and

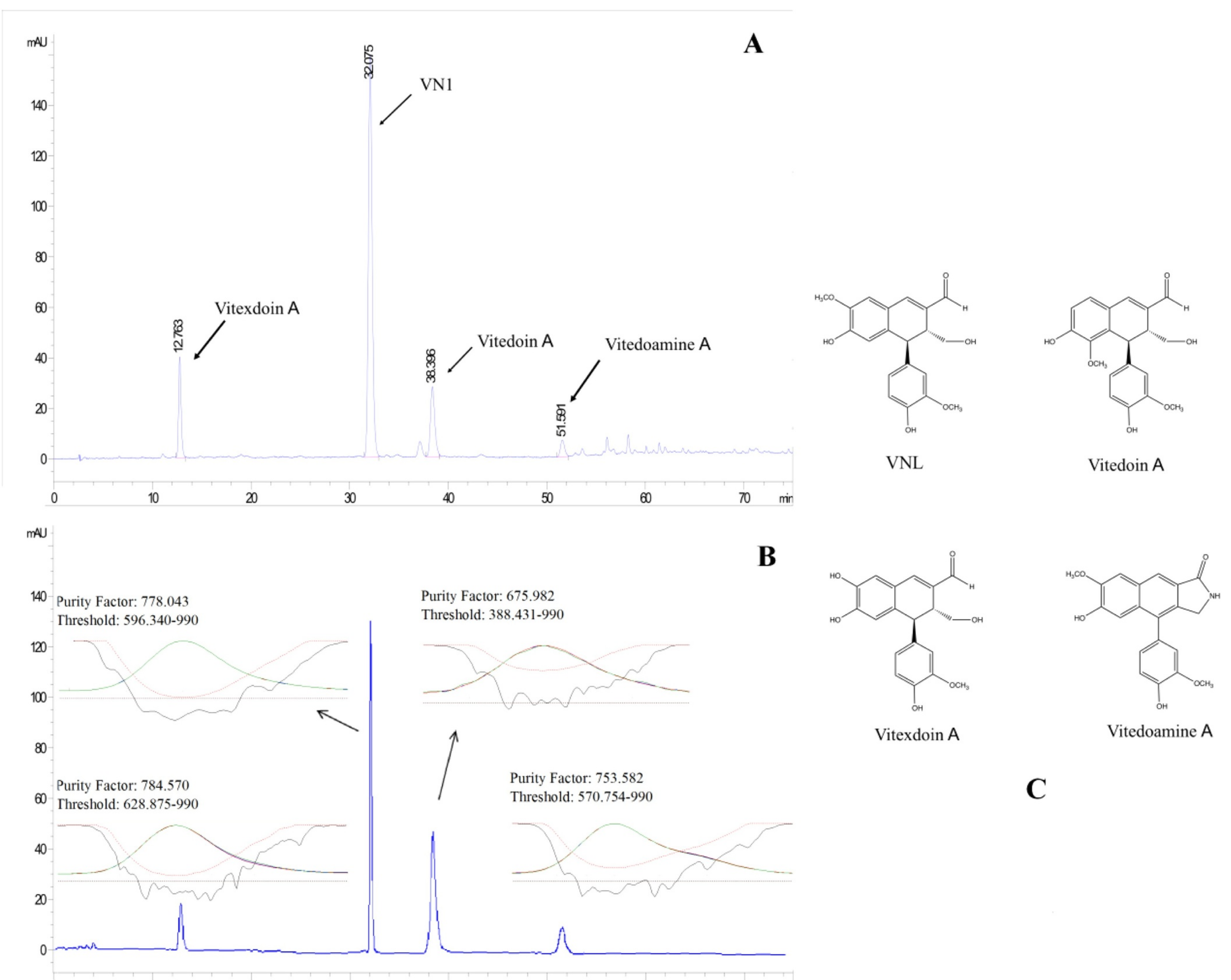

Fig 1. HPLC analysis of TOV (A), purity analysis (B) and chemical structures (C) of four major lignans, including VNL, Vitedoin A, Vitedoamine A and Vitexdoin A.

4 = the entire paw affected and maximal erythema and swelling (Kokkola et al., 2003). On the 51th day, the rats were sacrificed by decapitation under deep anesthesisa (pentobarbital sodium, 40 mg/kg, i.p.).

Index of thymus and spleen assay

At day 51 after administration, the rats were sacrificed after anesthesia. Then, the thymus and spleen were removed from animals, washed in ice-cold saline, patted dry and weighed (Sundaram et al., 2012). The index of thymus and spleen were expressed as the ratio of thymus and spleen wet weight *versus* body weight (mg/g), respectively (Zuo et al., 2014).

Histopathological studies

On day 51, all rats were sacrificed after anesthesia and keen joints were removed from rats for histopathological examination (n = 10). Then the tissues were fixed in 10% neutral buffered formalin and decalcified in EDTA for 30 day at 4 °C. After the process of paraffin embedding (Duwiejua et al., 1993), tissues sections (3 μm thick) were stained with haematoxylin-eosin and viewed under a light microscope for observation of histopathological changes (Connor et al., 1995).

Determination of the levels of serum IL-1β, IL-6, IL-8, IL-10, IL-17A, TNF-α, MMP-2, MMP-3 and MMP-9

Blood was obtained from rats on day 51 and allowed to clot for 1 h at room temperature, and subsequently centrifuged at 5000 g for 30 min at 4 °C. Serum was collected and frozen at –20 °C until assayed. Levels of serum IL-1β, IL-6, IL-8, IL-10, IL-17A, TNF-α, MMP-2, MMP-3 and MMP-9 were determined using commercially available ELISA kits according to the manufacturer's recommendations.

Western blotting

The synovial tissues were homogenized in RIPA lysis buffer containing phosphate inhibitors and protease inhibitor mixture. The proteins were extracted at 4 °C for 30 min and centrifuged two times at 10,000 g for 15 min each time at 4 °C. Supernatants were used for protein quantification by BCA Protein Assay kit and stored at –80 °C until used. Western blotting was performed as described previously (Peng et al., 2015). Briefly, approximately 40 μg of total protein was subjected to SDS-PAGE gel electrophoresis, transferred to PVDF membrane, then incubated with indicated primary antibodies (1:1000), followed by incubation with a HRP-conjugated secondary antibody (1:2000). Detection was performed with a Tannon 5200 chemiluminescence imaging system (Shanghai, China).

Xylene-induced ear edema in mice

The xylene-induced ear edema test with modifications was performed as previously described (Bitencourt et al., 2011). Briefly, ICR mice were divided randomly into five groups with ten animals in each as follows: group I (0.5% CMC−Na), group II (DXM, 1 mg/kg), group III (TOV 10 mg/kg), group IV (TOV 20 mg/kg) and group V (TOV 40 mg/kg). Sixty minutes after intragastrical injection of CMC−Na, DXM or TOV, 30 μl of xylene was applied to the anterior and posterior surfaces of the right ear. The left ear was considered as control. Thirty minutes after xylene administration, mice were sacrificed and both ears were removed. To evaluate the ear weight, 7 mm diameter ear punch biopsies were collected using a metal punch and individually weighted. The extent of ear edema was evaluated by the weight difference between the right and the left ear biopsies of the same animal. Inhibition level (%) was calculated according to the following equation:

Inhibition (%) = $[1 - Et/Ec] \times 100$, where Et = average edema in the treated group, and Ec = average edema of the control group.

Acetic acid-induced writhing in mice

Antinociceptive activities of TOV were evaluated by measuring the mice response to an intraperitoneal injection with acetic acid solution (0.7%, 0.2 ml/10 g body weight), which was presented as writhing (contraction of the abdominal muscles and full extension of both hind paws) (Fischer et al., 2008). ICR mice (n = 10) were intragastrically administrated with TOV (10, 20 and 40 mg/kg) or aspirin (10 mg/kg) sixty minutes before acetic acid injection. Control animals received a similar volume of saline. The number of writhing was recorded during the subsequent 25-min period. The antinociceptive activity was expressed as the inhibition percentage of abdominal writhing.

Acute toxicity test

Since no lethal dose was obtained for TOV in our preliminary test (data not shown), the maximum tolerated dose (MTD) in mice of TOV

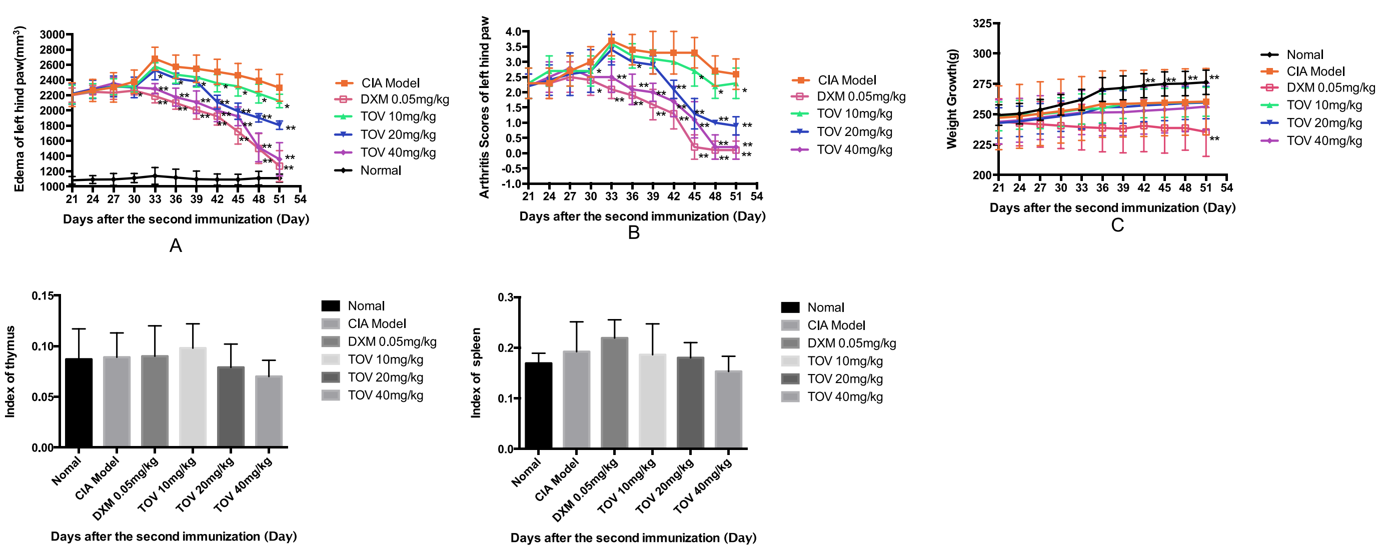

Fig 2. Effects of TOV on the paw edema (A), arthritis index score (B), weight growth (C) and index of spleen and thymus (D, E) of CIA rats. Values expressed as mean ± SD (n = 10). Rats were randomly divided into six groups after the second immunization, and then dexamethasone and TOV were orally administered from day 21 to 51. During the course of the experiment, the body weight and the left hind paw of rats were measured every 3 days. And the degree of arthritis was also observed every 3 days from the day of the second immunization. #p < 0.05, ##p < 0.01, ###p < 0.001, *vs.* normal control. *p < 0.05, **p < 0.01, ***p < 0.001, *vs.* CIA model control.

was examined. Twenty ICR mice fasted over night were randomly divided into two groups with ten mice per group of either sex. A single dose of TOV (16 g/kg) was administered orally twice to the mice in the test group during 24 h. Each animal in the control group was intragastrically given the same volume of 0.5% CMC−Na only. After administration, the mice were allowed free access to food and water, and observed under open-field conditions for 2 weeks. All external morphological, behavioral, neurologic, autonomic changes, number of dead animals and time to death, as well as some other toxic effects were recorded continuously over a period of 2 weeks. After 2 week observation, all animals were sacrificed after anesthesia and histopathologic examination of organ tissues were processed and evaluated.

Statistical analysis

Data are presented as mean ± standard deviation (SD) of indicated number of experiments. Statistical differences between groups were assessed by using the Student's test. Differences between different groups were considered significant when $p < 0.05$. All statistical analysis was performed using SPSS version 17.0 (SPSS Inc., Chicago, IL, USA).

Result

Chemical analysis of TOV

Based on our established HPLC chromatographic conditions, four main phenylnaphthalene-type lignans were identified in TOV (Zheng et al., 2009a) by comparing the retention times of individual peaks with those of the authentic reference standards, including VNL (22.14%), vitedoin A (24.54%), vitedoamine A (0.29%) and vitexdoin A (4.43%). The chemical structures of these four lignans are shown in Fig. 1.

Effects of TOV on the paw edema, arthritis index score, weight growth and index of spleen and thymus

To investigate the *in vivo* anti-arthritic effects of TOV, the efficacy of TOV was examined in collagen-induced arthritis in rats. Paw edema, arthritis index score, body weight and index of spleen and thymus were employed as parameters indicating arthritic symptoms in rats. Regarding the joint swelling in arthritic rat hind limb, the paw volume started to increase at day 21, reached a maximum level on day 33, and then decreased gradually. As shown in Fig. 2A, administration of TOV (10, 20 and 40 mg/kg day, ig) resulted in significant and dose-dependent amelioration of the left hind paw edema in CIA rats. Especially, TOV, at doses of 20 and 40 mg/kg, produced evident decrease ($p < 0.05$) in the left hind paw edema in CIA rats from day 33. TOV, at dose of 10 mg/kg, also exhibited notable efficacy ($p < 0.05$) in inhibiting the left hind paw edema of CIA rats from day 42. Meanwhile, treatment with DXM as the positive control could also significantly reduce the paw edema of CIA rats ($p < 0.05$).

After the second injection of CII and IFA, the similar trend as paw edema was also observed in the clinical arthritis score and obvious declines of this index emerged in both TOV and DXM groups. TOV treatment dose-dependently inhibited the aggravation of arthritic symptoms in CIA rats in terms of arthritis score and the maximum inhibition of TOV on the arthritis index score was observed at the dose of 40 mg/kg. The inhibitory effect of TOV at this dose was closely to that of the positive control (0.05 mg/kg DXM) in CIA rats (Fig. 2B).

As shown in Fig. 2C, rats in DXM treatment group showed a marked weight loss from day 30 to day 51 when compared with the CIA model rats ($p < 0.05$). However, animals treated with TOV, at 10, 20 and 40 mg/kg, did not show significant weight loss during the whole experiment compared with the model group.

In addition, when compared with the normal group, the index of spleen and thymus of rats in CIA model, DXM and TOV (10, 20 and 40 mg/kg) groups did not show obvious differences (Fig. 2D-E).

Effects of TOV on histopathological changes

To evaluate the anti-arthritic effects of TOV, samples of the hind paw ankle joints from different experimental groups were examined by H&E staining. We found that normal rats showed intact articular cartilage, normal joint space and synovial lining cells without infiltration and vascular proliferation (Fig. 3, Group I). The joints of CIA model rats presented cartilage damage and synovial hyperplasia with pronounced cell infiltration of macrophages and lymphocytes accompanied with fibroblasts proliferation (Fig. 3, Group II A-B). However, in TOV (40 mg/kg)-treated (Fig. 3, Group VI) and DXM-treated rats (Fig. 3, Group III), significant reduction of synovial inflammatory cell infiltration and fibroblasts proliferation was observed compared with the CIA

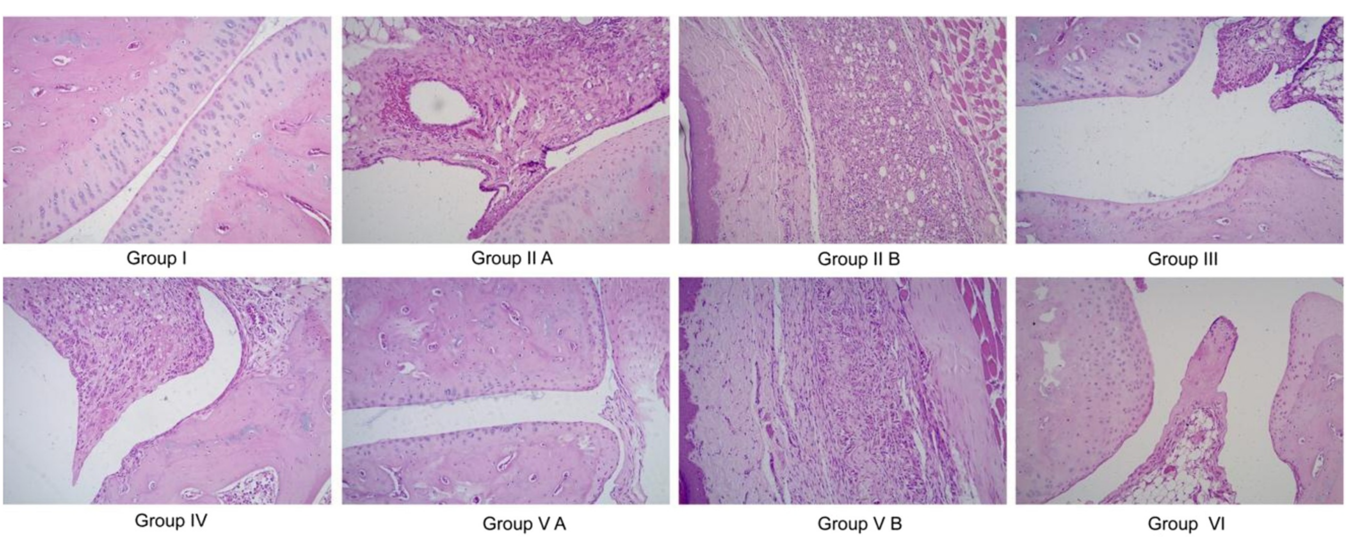

Fig 3. Effects of TOV on the histppathological changes in CIA rats. Group I – normal rats (H&E staining 100 ×), Group II (A, B) – CIA model rats, which shows cartilage damage and synovial hyperplasia with pronounced inflammatory cell infiltration (H&E staining 100 ×), Group III – CIA + DXM (0.05 mg/kg) (H&E staining 100 ×), Group IV – CIA + TOV (10 mg/kg) (H&E staining 100 ×), Group V(A, B) – CIA + TOV (20 mg/kg) (H&E staining 100 ×), Group VI – CIA + TOV (40 mg/kg) (H&E staining 100 ×).

model group. In addition, synovial hyperplasia and joint cartilage damage were scarcely detected and normal structure of joint cartilage was kept in the 40 mg/kg dose group of TOV, whereas synovial hyperplasia was still observed in DXM-treated rats. Although rats in the group treated with 20 mg/kg of TOV also exhibited synovial hyperplasia, marked decrease of inflammatory cell infiltration and fibroblasts proliferation was observed (Fig. 3, Group V A-B). The 10 mg/kg dose group of TOV revealed a slight decrease infiltration of macrophages and lymphocytes with mild injure in articular cartilage, but the alterations in synovial hyperplasia were not obvious (Fig. 3, Group IV).

Effects of TOV on the expressions of IL-1β, IL-6, IL-8, IL-10, IL-17A, TNF-α, MMP-2, MMP-3 and MMP-9 in serum

The expression levels of IL-1β, IL-6, IL-8, IL-10, IL-17A, TNF-α, MMP-2, MMP-3 and MMP-9 in the serum of CIA rats were shown in Fig. 4. Marked increase in the expressions of IL-1β ($p < 0.01$), IL-6 ($p < 0.01$), IL-8 ($p < 0.01$), IL-17A ($p < 0.01$), TNF-α ($p < 0.01$), MMP-3 ($p < 0.01$) and MMP-9 ($p < 0.01$) in rat serum were observed in CIA model group, while the level of IL-10 ($p < 0.01$) was declined, compared to that of the normal group. Both TOV (20 and 40 mg/kg) and DXM (0.05 mg/kg) showed potent inhibitory effects on the production of IL-1β, IL-6, IL-8, IL-17A, TNF-α, MMP-3 and MMP-9. In addition, the level of IL-10 was significantly increased in TOV (20 and 40 mg/kg) and DXM (0.05 mg/kg) group. Administration of 10 mg/kg of TOV only displayed a significant impact on the expression of IL-8, IL-17A, TNF-α and MMP-3.

Effects of TOV on the expressions of COX-2, iNOS, IκB and p-IκB in synovial tissues

As illustrated in Fig. 5, the expressions of COX-2, iNOS, *p*-IκB were significantly suppressed by TOV in a dose-dependent manner ($p < 0.05$), while an increase in total IκB expression was observed in synovial tissues when compared with the CIA model group. The results revealed that the low dose group of TOV (10 mg/kg) exhibited a mild inhibitory effect on the expressions of COX-2 and iNOS ($p < 0.05$), while the higher dose groups of TOV (20 mg/kg and 40 mg/kg) and DXM (0.05 mg/kg) significantly suppressed ($p < 0.01$) the production of COX-2 and iNOS. In addition, TOV at 10, 20 and 40 mg/kg and DXM at 0.05 mg/kg showed an evident impact on the expressions of IκB and *p*-IκB in synovial tissues, indicating the inhibitory effects of these drugs on NF-κB pathway.

Effects of TOV on xylene-induced ear edema in mice

As Fig. 6A shown, treatment with TOV (10, 20 and 40 mg/kg) significantly inhibited the xylene-induced ear edema in a dose-dependent manner and the inhibition rates were 22.66%, 32.23% and 35.63%, respectively, as compared to the vehicle control. The positive control DXM (1 mg/kg) also significantly suppressed the xylene-induced ear edema by 45.73% when compared to the vehicle control.

Effects of TOV on acetic acid-induced writhing in mice

The results of acetic acid-induced writhing test were summarized in Fig. 6B. Aspirin, at dose of 10 mg/kg, was used as positive control and significantly diminished the writhing response induced by acetic acid injection, with an inhibition rate of 35.76%. TOV (10, 20 and 40 mg/kg), in a dose dependent manner, significantly inhibited the number of writhes compared to normal control and the inhibition rates were 22.16%, 30.26% and 42.42%, respectively.

Acute toxicity of TOV

In the maximum tolerated dose (MTD) test, no obvious adverse effects were observed in mice treated with 16 g/kg of TOV during the 2-week experimental period and the clinical appearance of the treatment group was similar to that of the control group. Changes of the animal body weight throughout the test were shown in Table 1. There was no statistical difference in the body weight and weight gain between the treated and control groups of the same sex. Ratios between the organ weights (liver, spleen, kidneys and hearts) and the body weight in TOV-treated animals were also similar to those of the control (Table 2).

Discussion

RA is a chronic autoimmune and inflammatory disease that primarily affects the synovial joints, which is characterized by cytokine-mediated inflammation of the synovial lining of the joints, leading to erosions of the cartilage and bone, and even joint deformity as well as dysfunction and ultimately to disability (Doeglas et al., 1995; Wong et al., 2010). About 1% of the world's population is affected by RA, and women are three times more likely than men to develop RA between the age of 35 and 50 years old (Chan and Carter, 2010). However, current therapy such as NSAIDs, DMARDs and glucocorticoids, while

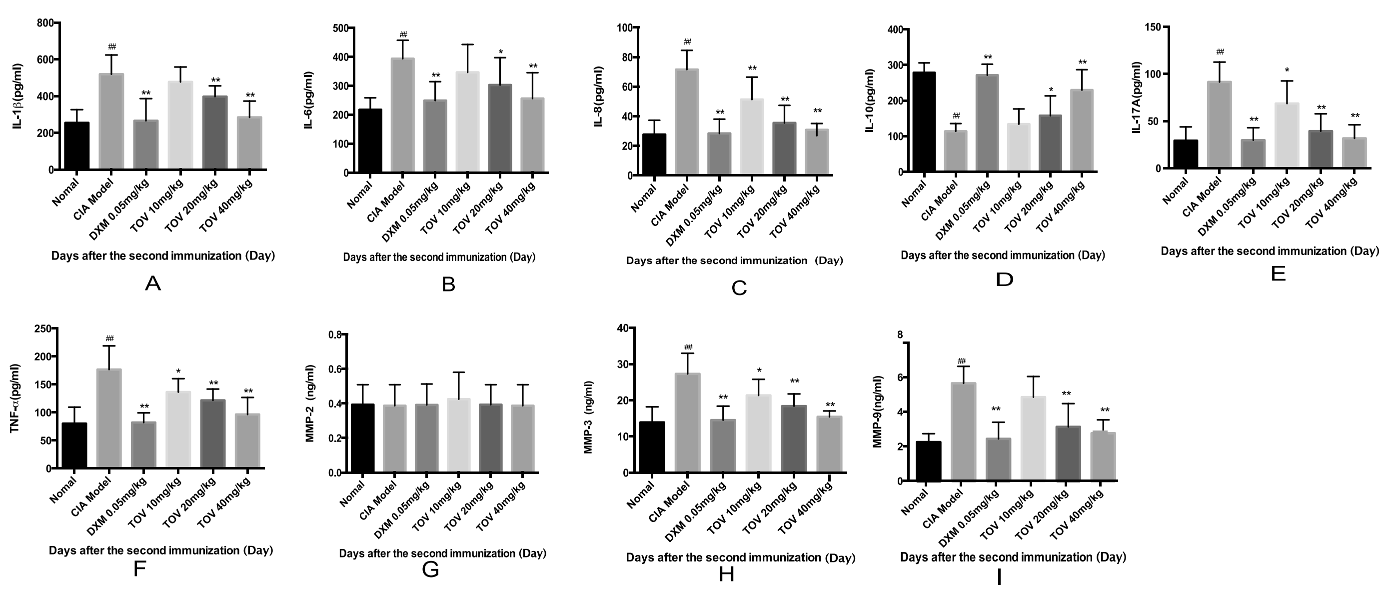

Fig 4. Effects of TOV on collagen-induced expression of IL-1ββ (A), IL-6 (B), IL-8 (C), IL-10 (D), IL-17A(E), TNF-α (F), MMP-2 (G), MMP-3 (H) and MMP-9 (I) in serum. The results are expressed as mean ± SD ($n = 10$) in each group. #$p < 0.05$, ##$p < 0.01$, ###$p < 0.001$, *vs.* normal control. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, *vs.* CIA model control.

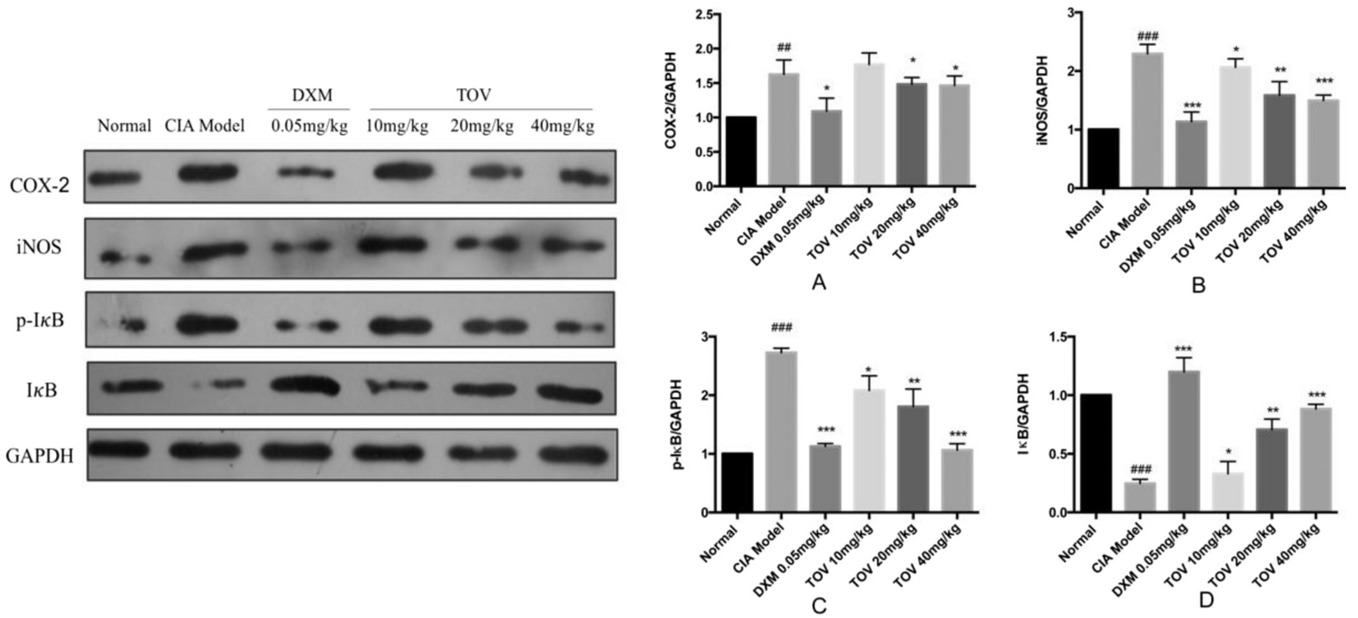

Fig 5. Effects of TOV on collagen-induced expression of COX-2, iNOS, IκB and p-IκB in synovial tissues by western blotting. (A) The immunoblot bands of COX-2, iNOS, IκB and p-IκB. (B) Relative level of COX-2 (normalized to GAPDH). (C) Relative level of iNOS (normalized to GAPDH). (D) Relative level of IκB (normalized to GAPDH). (E) Relative level of p-IκB (normalized to GAPDH). The results are shown as the as mean MMP-2 (G), MMP-3 (H) and MMP-9 (I) in serum. The results are expressed as mean ± SD of at least three independent experiments. $\#p < 0.05$, $\#\#p < 0.01$, $\#\#\#p < 0.001$, *vs.* normal control. $*p < 0.05$, $**p < 0.01$, $***p < 0.001$, *vs.* CIA model control.

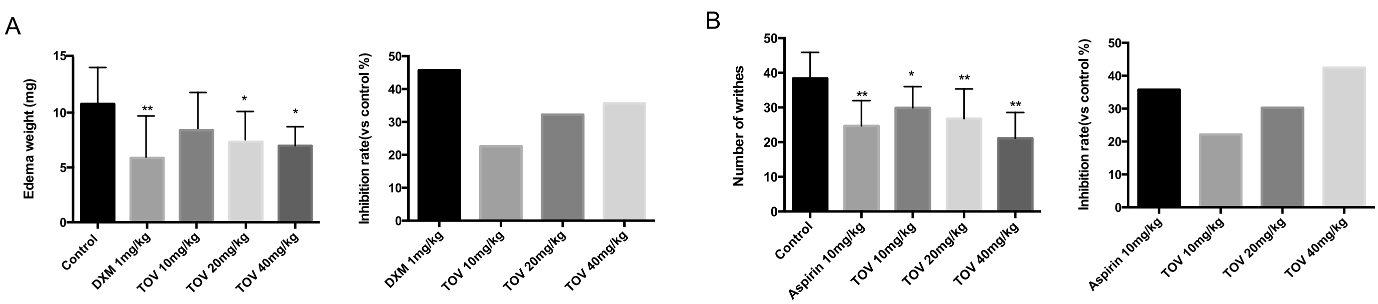

Fig 6. Effects of TOV on xylene-induced ear edema in mice and effects of TOV on acetic acid-induced writhing in mice. (A) The edema weight and inhibition (%) of ear edema formation (compared to control group). The control group was administered with 0.5% CMC−Na as a vehicle. (B) The number of writhes and inhibition (%) of the number of writhes. The control group was administered with saline as a vehicle. $*p < 0.05$, $**p < 0.01$, $***p < 0.001$, *vs.* control.

providing symptomatic relief, does not alter the fundamental mechanisms of the disease process or its progression (Roubille et al., 2015; Smolen and Aletaha, 2015). Recently, pro-inflammatory cytokine blockers, such as IL-6 inhibitors and TNF-α inhibitors, play an important role in the treatment of RA (Furst and Emery, 2014; Siebert et al., 2015). However, these drugs achieve disease remission in only a minority of patients due to infectious adverse events, rebound of symptoms, short half-lives, or high treatment costs (Huscher et al., 2015; Lahiri and Dixon, 2015; Zampeli et al., 2015). Thus, it naturally draws great attention of extensive experimental researchers and clinical observations to search for new therapeutic drugs with high efficiency and lower toxicity for RA from the natural sources.

Vitex negundo, an important medicinal plant in genus *Vitex*, is widely distributed in tropical to temperate regions, native of South Asia, China, Japan, Indonesia, South America and East Africa (Zheng et al., 2010). This medicinal plant possesses versatile pharmacological activities,

Table 1
Effect of oral treatment of TOV on body weight of mice.

Group	Total body weight (g) 1d	3d	5d	7d	10d	14d
Male						
Control	18.0 ± 0.4	20.8 ± 0.4	22.1 ± 0.5	23.3 ± 0.8	25.5 ± 0.8	28.2 ± 0.8
TOV(16 g/kg)	18.6 ± 0.6	21.5 ± 0.2	22.7 ± 0.4	23.5 ± 0.2	25.1 ± 0.3	27.2 ± 0.2
Female						
Control	18.6 ± 0.3	21.6 ± 0.5	23.1 ± 0.5	24.5 ± 0.3	26.5 ± 0.5	28.5 ± 0.4
TOV(16 g/kg)	19.2 ± 0.3	22.2 ± 0.7	23.4 ± 0.2	24.6 ± 0.2	26.2 ± 0.5	28.7 ± 0.5

Data are expressed as mean ± SD ($n = 10$). The control group was administered with 0.5% CMC−Na as a vehicle. $*p < 0.05$, $**p < 0.01$, *vs.* control group.

Table 2
Effect of oral treatment of TOV on ratio of organ weight to body weight of mice.

Group	Ratio of organ to body weight (%)			
	Liver	Heart	Kindeys	Spleen
Male				
Control	5.1 ± 0.4	0.5 ± 0.2	1.3 ± 0.3	0.5 ± 0.2
TOV(16 g/kg)	4.8 ± 0.3	0.4 ± 0.1	1.2 ± 0.2	0.4 ± 0.2
Female				
Control	5.3 ± 0.4	0.4 ± 0.2	1.3 ± 0.2	0.6 ± 0.2
TOV(16 g/kg)	5.2 ± 0.1	0.3 ± 0.2	1.2 ± 0.3	0.5 ± 0.1

Data are expressed as the mean ± SD (n = 10). The control group was administered with 0.5% CMC–Na as a vehicle. *p < 0.05, **p < 0.01, *vs.* control group.

such as anti-inflammatory (Chawla et al., 1992; Zheng et al., 2009a; Zheng et al., 2010), analgesic (Zheng et al., 2009a), antitumor (Zhou et al., 2009) and antioxidant properties (Zheng et al., 2009b). Especially, the seeds of this plant have been widely used in traditional Chinese medicine for anti-inflammatory and anti-arthritic purpose (Chawla et al., 1992; Zheng et al., 2014b; Zheng et al., 2009a; Zheng et al., 2010). In our previous studies, we found that *V. negundo* seeds possessed rich lignans metabolites, especially phenylnaphthalene-type lignans (Zheng et al., 2009a,b; Zheng et al., 2014a), which were reported to display significant anti-inflammatory activities. So we deduced that the total lignans of *V. negundo* seeds (TOV) may play an important role in the treatment of arthritis. This study was therefore conducted to prepare the TOV from *V. negundo* seeds and investigate its anti-arthritic effects on collagen-induced arthritis (CIA) in rats as well as its possible mechanisms.

In our study, TOV was prepared by combined macroporous resin and polyamide column chromatography, and constituents of TOV were analyzed by HPLC, which revealed the presence of four major phenylnaphthalene-type lignans, including VNL (22.14%), vitedoin A (24.54%), vitedoamine A (0.29%) and vitexdoin A (4.43%). The content of those measurable constituents was more than 50%. VNL, one principal lignan in TOV, was proved to be an effective anti-inflammatory agent (Zheng et al., 2009a). Vitedoin A was a positional isomer of VNL. Vitedoamine A and vitexdoin A also possessed the similar skeleton as VNL. Thus, we suggested that this type of lignans in *V. negundo* seeds may be responsible for the reported anti-arthritic activity of *V. negundo* seeds, which merits further studies regarding the efficacy and mechanism of action.

CIA model in rats was established by immunization with chicken type II collagen. Paw edema and arthritis index are important indices for reflecting the anti-arthritic activity of tested drugs and employed here to evaluate the anti-arthritic activity of TOV. Our results revealed that administration of TOV resulted in significant, dose-dependent amelioration of the hind paw edema and decrease in arthritis index when compared with the CIA model group. In addition, TOV treatment did not show obvious influence on the body weight growth of the CIA rats. However, DXM, as the positive drug, significantly decreased the growth of the animal body weight from day 30 to 51 when compared with the CIA model rats, which indicated that TOV may be safer than DXM in the treatment of RA.

Histopathological examinations of synovial tissues revealed that pathological characteristics observed in CIA rat model were in accordance with the pathological progression of RA. In CIA model animals, we observed cartilage damage and synovial hyperplasia with pronounced cell infiltration of macrophages and lymphocytes accompanied with fibroblasts proliferation. However, our study revealed that TOV not only suppressed inflammatory response, but also alleviated cartilage damage in CIA rats, thus may be one favorable candidate drug for RA treatment.

It is well known that the inflammatory melieu in the synovial compartment is regulated by a complex cytokine and chemokine network (Morino and Robecchi, 2010). In RA, IL-1β, the predominant extracellular form of IL-1, has been regarded as an important pro-inflammatory cytokine in the pathogenesis of RA mainly causing bone and cartilage destruction, while TNF-α appears to be primarily responsible for driving inflammation (Karatay et al., 2004). IL-6, a key cytokine in RA, can not only contribute to the production of matrix metalloproteinases (MMPs) of synoviacytes, but also induce the differentiation and formation of osteoclasts (Lai et al., 2017). IL-8, a chemokine in RA, is involved in neovascularization and may play a role in neutrophil accumulation within the synovial fluid (Feldmann et al., 1996). IL-10, in contrast, is a potent anti-inflammatory cytokine (Mazza et al., 2010), which has been shown to regulate endogenous proinflammatory cytokine production in RA synovial tissue (Mitchell et al., 2008). IL-17A also plays a crucial role in RA, which promotes the release of additional proinflammatory cytokines, chemokines and matrix metalloproteinases, as well as facilitating osteoclast activation and angiogenesis. Our results showed that the levels of serum IL-1β, IL-6, IL-8, IL-17A and TNF-α rose significantly in CIA model rats, while a marked decrease of IL-10 was observed. However, administration of TOV markedly attenuated the expressions of IL-1β, IL-6, IL-8, IL-17A and TNF-α in the serum of CIA rats and increased the level of IL-10. In addition, matrix metalloproteinases (MMPs), a family of proteolytic enzymes, are also involved in RA, which are largely responsible for the irreversible destruction of cartilage, bone and tendons in the joints. We also found that TOV could significantly reduce the expressions of MMP-3 and MMP-9, which suggests that the anti-arthritic effect of TOV may not only involve cytokines and chemokines expression regulation, such as down-regulating the levels of IL-1β, IL-6, IL-8, IL-17A and TNF-α and up-regulating the expression of IL-10, but also interfere the expressions of matrix metalloproteinases such as MMP-3 and MMP-9.

Cyclooxygenase-2 (COX-2), an important enzyme in prostaglandins (PG) biosynthesis, is an lipid mediator that are produced at high levels in inflamed tissues such as rheumatoid synovium (Davies et al., 1984; Martel-Pelletier et al., 2003; Masferrer et al., 1994; Bombardieri et al., 1981). NO is an inflammatory mediator that is readily detected at various sites of inflammation. Up-regulation of inducible isoform of nitric oxide synthase (iNOS) generates excessive NO production during the course of a variety of diseases, including RA (Amin et al., 1999) and other inflammatory disorders. Our results exhibited that high levels of COX-2 and iNOS were expressed in synovial tissues of CIA rat, while significant decrease was observed in TOV treated animals. NF-κB is one of the pivotal regulators of pro-inflammatory gene expression and it induces the transcription of pro-inflammatory cytokines, chemokines, adhesion moleculars, MMPs, COX-2 and iNOS (Baeuerle and Baichwal, 1997; Tak and Firestein, 2001), which is highly activated at sites of inflammation in diverse diseases, including rheumatoid arthritis and cancers (Firestein, 2004). Whereas, activation of NF-κB is facilitated by phosphorylation and subsequent dissociation of its inhibitor, IκB (Mcintyre et al., 2003). Therefore, we further investigated the effects of TOV on the expression of IκB and clearly found that TOV significantly suppressed the phosphorylation and degradation of IκB, suggesting that TOV regulated the levels of serum IL-1β, IL-6, IL-8, IL-10, IL-17A, TNF-α, MMP-2 and MMP-3 as well as the expression of COX-2 and iNOS probably through suppressing the phosphorylation and degradation of IκB.

In addition, we found that TOV exhibited significant anti-inflammatory effects in xylene induced ear edema in mice, an acute inflammation model, and also displayed notable analgesic activity in acetic acid-induced writhing in mice. In the acute toxicity test, no lethal dose was observed and the MTD of TOV was determined to be 16 g/kg, the crude drug amount of which was about 6400 times that of the clinical dosage of *V. negundo* seeds, thus indicating the high safety for TOV.

In conclusion, our study showed that TOV was effective on CII-induced arthritis in rats, as evidenced by both signs and pathology scores. The anti-arthritic activity of TOV observed can be attributed to its ability in down-regulating the levels of IL-1β, IL-6, IL-8, IL-17A, TNF-α, MMP-2 and MMP-9 and up-regulating the expression of IL-10 as well as decreasing the expression of inflammatory enzymes COX-2 and iNOS, probably through the inhibition of the phosphorylation and degradation of IκB. Combined with its high safety revealed by the acute toxicity test, we suggest that TOV, a bioactive fraction from *V. negundo* seeds, can be used as a potent and promising strategy for the treatment of RA, which merited further studies as regards to fully elucidate the anti-RA components, synergistic effects and mechanisms of action.

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Conflict of interest

We wish to confirm that there are no known conflicts of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome.

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